Quantum-Inspired Raman Scattering-based Noninvasive Glucose-Level Classification through Machine Learning

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Abstract. Noninvasive glucose monitoring is continuously posing a significant challenge because Raman signals are intrinsically weak, noisy, and extremely sensitive to physiological variability. Despite the fact that Raman spectroscopy provides molecule-specific information, extraction of discriminative glucose-related patterns from high-dimensional spectra has proved challenging, especially with a limited number of real samples. To address these challenges, this paper proposes a quantum-inspired Raman scattering-based framework for noninvasive glucose-level classification. The framework consists of logarithmic transformation, gradient-based spectral enhancement, mathematical modeling for class-specific synthetic Raman spectrum generation, and quantum-inspired feature engineering that includes Principal Component Analysis (PCA) with angle encoding to generate a qubit-ready eight-dimensional representation. A Stacking Ensemble classifier is used as the final model to leverage the benefits of different base learners. Through experimental evaluation using a leakage-free stratified 5-fold cross-validation in which the PCA projection and angle-encoding scaler are refit within each training fold, the proposed framework achieved an accuracy of $90.14 \pm 1.14\%$, remaining highly competitive with representative Raman-based and quantum-inspired glucose classification algorithms while exhibiting the most consistent performance among the top-performing models. These results indicate that the proposed quantum-inspired framework is an effective and reliable approach for glucose-level classification from limited Raman data and provides a foundation for future hybrid quantum-classical healthcare applications.

Keywords: Raman Spectroscopy · Quantum-Inspired Learning · Machine Learning · Noninvasive Glucose Monitoring · PCA · Synthetic Data Generation · Pattern Classification.

1 Introduction

Diabetes mellitus is one of the world's leading chronic diseases, currently impacting almost half a billion people and expected to significantly grow in the coming years [8, 17]. Reliable monitoring of blood sugar levels is one of the key aspects of disease treatment. Traditional methods of glucose level detection are usually invasive, such as finger-pricking tests or subcutaneous biosensors, which bring discomfort, risk of infection, and cannot be used for long-term monitoring [5, 20]. Thus, there is a necessity to develop non-invasive ways of glucose level measurement, which would not involve any kind of skin penetration, yet still provide accurate estimation of the sugar content in the blood.

Among existing approaches to non-invasive sensing, Raman spectroscopy seems like the most promising one due to its ability to detect vibrational modes of individual molecules, which enables distinction of glucose-specific spectra from other biological components in tissue [9, 11]. However, Raman scattering spectra acquired from skin or biofluids are typically weak, susceptible to background fluorescence and instrument noise, as well as highly dependent on inter-subject physiological variance [2, 16]. Therefore, extraction of a stable glucose-related pattern from a set of Raman scattering data represents a difficult signal processing and machine learning task.

Machine learning models, applied directly to raw multidimensional Raman scattering spectra, typically have difficulties capturing the subtle nonlinear relationships between glucose-specific classes, which has led to recent research focusing on ensemble or transfer learning classifiers for glucose concentration estimation based on Raman scattering data [10, 19]. Moreover, in cases of small training dataset, which is usually characteristic of early-stage biomedical sensing research, overfitting risk becomes especially high [6]. Thus, there is an evident gap in the current knowledge on non-invasive glucose monitoring methods: a necessity for a learning algorithm, which (i) could handle a small number of actual Raman scattering samples, (ii) enhance and stabilize glucose-sensitive spectral information and (iii) could present the results in a structured way compatible with future quantum computing paradigm [12, 18], without the need of quantum processors, which are not available to everyone yet [7, 15].

This paper introduces a quantum-inspired approach for noninvasive glucose level classification based on Raman scattering, combining spectral preprocessing, data-driven synthetic augmentation, and a PCA-based quantum feature encoding step. The resulting quantum-ready representation is benchmarked across a broad set of classical and ensemble-based classifiers to identify the most effective learning strategy, while laying a practical foundation for future work involving genuine quantum circuits and quantum sensing hardware. The main contributions of this paper are summarized as follows.

- We propose a quantum-inspired Raman scattering-based framework for noninvasive glucose-level classification, with a Stacking Ensemble classifier identified as offering the most consistent learning strategy among several closely competitive classifiers.

- We formulate a mathematical model for class-specific synthetic Raman spectrum generation using statistical characteristics estimated from real training data to address the limited-sample-size challenge of Raman-based glucose datasets.
- We introduce a quantum feature engineering pipeline that transforms high-dimensional Raman spectra into an eight-dimensional qubit-ready representation using PCA and angle encoding.
- We comprehensively evaluate the proposed framework using multiple performance metrics and have demonstrated its effectiveness compared with representative Raman-based and quantum-inspired glucose classification methods.

Organization: The rest of this paper is organized as follows. In Section 2, we review the literature relevant to non-invasive glucose methods and Raman Spectroscopy sensing techniques. In Section 3, we describe the proposed technique, which includes the quantum ready feature engineering pipeline. The results of the performance evaluation are presented in Section 4, where we benchmark our proposed system against selected previous works. We conclude the paper in Section 5.

2 Related Works

Noninvasive glucose sensing has been a focus of intensive research for the past decade due to pain and risks of infection associated with invasive and minimally invasive glucose measurements. Optical methods of sensing are widely used among many others for the development of noninvasive glucose sensing technologies, and Raman spectroscopy is one of those that draws considerable interest due to the possibility of molecule-specific vibrational fingerprints that could allow isolation of glucose-related spectral signatures among other tissue components [11]. However, the task of turning raw Raman spectral data into clinically meaningful glucose estimates is complicated due to signal-to-noise ratio, background interference, and difficulties in construction of a good prediction model based on usually limited number of samples. This section is dedicated to the review of recent developments in three areas relevant to this challenge: noninvasive glucose sensing systems using Raman spectroscopy, machine learning algorithms for spectral classification in small sample datasets, and quantum and quantum-inspired machine learning algorithms for healthcare applications.

There has been some recent progress in the field of noninvasive Raman spectroscopy-based glucose sensing systems. Rothenbühler et al. [16] performed a prospective clinical pilot study using a Raman spectroscopy sensor placed at the wrist coupled with a machine learning algorithm, reaching a mean absolute relative difference of 14.5% without subject-specific calibration, thus proving the applicability of glucose estimation using Raman spectroscopy coupled with the right machine learning model. In order to address the issue of hardware miniaturization of Raman spectroscopy based sensor, Bresci et al. [2]developed a band-pass Raman spectroscopy technique using off-axis near-infrared illumination, which allowed to develop a continuous glucose monitoring system based on portable device tested in vitro on tissue phantoms and in vivo on human skin. Liu et al. [10] studied in vivo Raman spectroscopy as a means of transcutaneous glucose monitoring with 785 nm laser system, both in animals and humans, demonstrating strong correlation of acquired spectral information with the level of blood glucose. Yang et al. [19] developed a vein visualization-guided confocal Raman spectroscopy system with heterogenous ensemble learning model, which allowed to reach 92% classification accuracy in pre- and postprandial glucose states after transfer learning in order to connect synthetic and in vivo transcutaneous data. Similarly, Li et al. [9] demonstrated effectiveness of Principal Component Analysis combined with backpropagation artificial neural network in reducing Raman spectral data dimensionality without losing glucose discrimination information.

Alongside spectroscopic techniques, Aloraynan et al. [1] proved that the combination of machine learning models and mid-infrared photoacoustic spectroscopy may be used for achieving accurate single-wavelength glucose detection. This tendency reflects the general trend of integrating different optical sensing techniques and machine learning models in order to estimate glucose level without blood testing. One of the recurring problems in this sphere is the relatively small size of the actual datasets from clinical measurements obtained using Raman spectroscopy, which decreases the reliability of classifiers trained based on these data. The same problem was addressed by Flanagan and Glavin [6] in the context of comparing statistical data-synthesis techniques and variational autoencoder-based generation for improving classification accuracy of small-size Raman spectral datasets. This study gives arguments in favor of implementing data-driven synthetic augmentation strategies, such as class-conditional statistical resampling technique, in situations where the number of actual samples is limited due to high costs of collecting the real Raman spectra.

At the same time, quantum and quantum-inspired machine learning has become one of the most popular trends in biomedical data processing. Maheshwari et al. [12] made a systematic review of the existing applications of quantum machine learning in the biomedical area. Based on their work, one may say that the variational quantum classifiers and quantum support vector machines became the most widespread architecture implementations for feature-based medical datasets. Ullah et al. [18] made another systematic review of quantum machine learning in healthcare area, paying attention to both growth of the number of publications and still existing difference between simulated and hardware-validated quantum algorithms. More recently, Gupta et al. [7] carried out a systematic review of quantum machine learning for digital health from 2015 to 2024, pointing out that there is no evidence of an empirical superiority of quantum algorithms over classical approaches in decision-making in healthcare area and that almost all studies use simulated quantum hardware instead of real one.

As for technical aspects of the problem, Chen et al. [3] suggested combined and splitting approaches to data encoding in the case of variational quantum classifiers working with complex-valued data, utilizing PCA for reducing the dimensionality of data and then applying an angle-based quantum encoding approach. It is consistent with the general approach to reducing the dimensionality of high-dimensional classical features in order to match the number of qubits and using rotation-based quantum gates.

To sum up, all of the presented papers reflect the potential of Raman spectroscopy, data augmentation and quantum machine learning for developing non-invasive glucose-level classification approaches. However, no existing research combines these components into one coherent framework and systematically compares multiple machine learning classifiers. This deficiency is the motivation for the development of the proposed framework.

3 Proposed Methodology

This work presents a novel framework based on quantum-inspired Raman scattering that can classify the levels of glucose non-invasively. As shown in Fig. 1, the framework comprises six sequential stages, which include Raman spectra collection, generation of class-specific synthetic spectra, spectral pre-processing, quantum-inspired feature engineering, machine learning models, and Model Training and Performance Evaluation. Firstly, the Raman spectra are augmented through the class-specific statistical model in order to overcome the limited sample problem. Thereafter, the spectra are improved and converted into an eight-dimensional qubit-ready representation using PCA and Angle encoding. Lastly, various machine learning models are evaluated, while the proposed Stacking Ensemble model is used as the classification model. The over- all framework provides a systematic pipeline for robust glucose-level classification while establishing a practical foundation for future quantum-inspired healthcare applications.

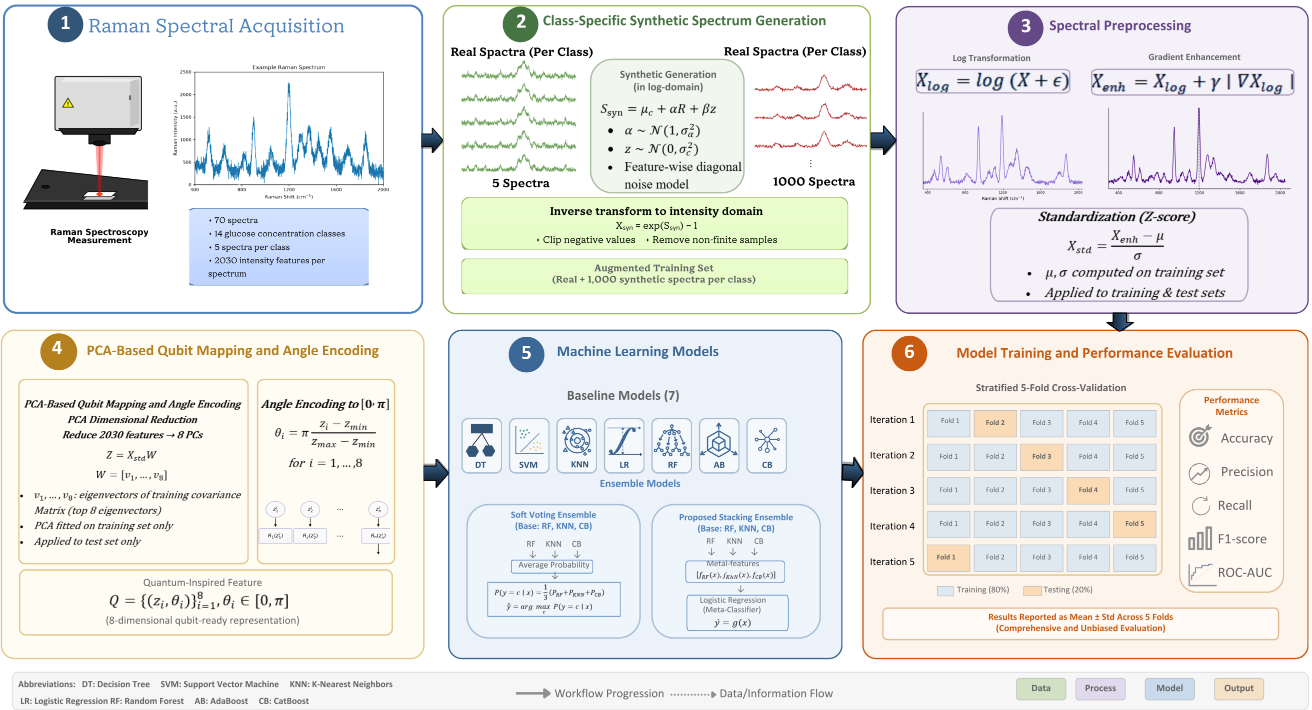

Fig. 1. Overall workflow of the proposed quantum-inspired Raman scattering-based glucose-level classification framework, illustrating Raman spectral acquisition, class-specific synthetic spectrum generation, spectral preprocessing, PCA-based qubit mapping with angle encoding, machine learning and ensemble classification, and stratified 5-fold performance evaluation.

3.1 Raman Spectral Acquisition

This study utilizes a publicly available Raman spectroscopy dataset for noninvasive glucose-level classification, available from the Mendeley Data repository [13]. The dataset consists of 70 Raman spectra distributed across 14 glucose concentration classes (M50, M52, M54, M55, M56, M58, M60, M62, M64, M65, M66, M68, M70, and M75), with five spectra per class. Each spectrum contains 2,030 Raman intensity measurements representing the recorded spectral response. The dataset serves as the basis for developing and evaluating the proposed quantum-inspired Raman scattering-based glucose-level classification framework.

3.2 Class-Specific Synthetic Spectrum Generation

Raman intensities are first stabilized through a logarithmic transformation,

$$X_{\log} = \log(X + \epsilon), \quad \epsilon > 0. \tag{1}$$

For each class c, the log-domain mean spectrum μ_c is computed from the training spectra of that class. A residual is computed from a randomly selected real spectrum, $R = S_{\text{real}} - \mu_c$, and a synthetic log-spectrum is generated as

$$S_{\text{syn}} = \mu_c + \alpha R + \beta z, \tag{2}$$

where $\alpha \sim \mathcal{N}(1, \sigma_\alpha)$ preserves class-consistent structure, and $z \sim \mathcal{N}(0, \sigma_c^2)$ is feature-specific Gaussian noise scaled by β, with σ_α and σ_c estimated per-feature from the real training spectra (a diagonal noise model is used in place of a full covariance matrix, since the number of training samples per class is too small to estimate a stable full covariance structure). This statistical, class-conditional approach to synthetic spectrum generation is consistent with established strategies for augmenting small-sample Raman spectral datasets [6]. The synthetic spectrum is mapped back to the intensity domain via $X_{\text{syn}} = \exp(S_{\text{syn}}) - 1$, followed by clipping of negative values and removal of non-finite samples. This procedure generates 1,000 synthetic spectra for each of the 14 glucose concentration classes, resulting in an augmented dataset of 14,000 Raman spectra. The augmented dataset is subsequently used for spectral preprocessing, quantum-inspired feature engineering, and model development [4].

3.3 Spectral Preprocessing

The augmented training spectra (real and synthetic) are further enhanced to emphasize glucose-sensitive spectral variations using a gradient-based term,

$$X_{\text{enh}} = X_{\log} + \gamma\, |\nabla X_{\log}|, \tag{3}$$

where γ is a fixed weighting coefficient. The enhanced spectra are subsequently standardized prior to dimensionality reduction.

3.4 PCA-Based Qubit Mapping and Angle Encoding

Principal Component Analysis reduces the enhanced 2,030-dimensional spectra to eight principal components,

$$Z = X_{\text{enh}} W, \quad W = [v_1, v_2, \ldots, v_8], \tag{4}$$

where $v_1, \ldots, v_8$ are the eigenvectors of the training-set covariance matrix corresponding to its eight largest eigenvalues. The PCA projection is fitted exclusively on $\mathcal{D}_{\text{train}}$ and subsequently applied to $\mathcal{D}_{\text{test}}$. Each component is then normalized into $[0, \pi]$,

$$z_i' = \pi\, \frac{z_i - z_{\min}}{z_{\max} - z_{\min}}, \tag{5}$$

so that each feature can be directly interpreted as a rotation angle for a parameterized quantum gate (e.g., R_x, R_y, R_z), following the standard PCA-plus-angle-encoding strategy used for preparing classical features for qubit-based learning [3]. The resulting eight-dimensional quantum-inspired dataset is denoted $\mathcal{Q} = \{(\mathbf{z}_i, y_i)\}_{i=1}^{N_q}$, with $\mathbf{z}_i \in [0, \pi]^8$, and forms the input to the downstream classifiers. However, the rationale behind angle encoding is not to make any claims about classification superiority compared to the traditional PCA representation, but to create an inherently bounded and interpretable representation which would directly fit to the rotation-based circuit. While the dimensionality reduction by PCA focuses on compression of the original high-dimensional spectrum, the angle mapping will transform each one of the retained components to the domain of valid angles of rotation of the parameterized quantum gate. As a result, the eight-dimensional representation would be usable for the classical classifiers utilized in this research and easily transferable to future quantum-based algorithms without modification of the representation. At the same time, the transformation brings all the retained components to a single bounded scale, thus eliminating the scale difference between the dimensions of PCA.

3.5 Machine Learning Models

To evaluate the effectiveness of the proposed quantum-inspired Raman feature representation, seven widely used supervised machine learning classifiers are employed as baseline models, namely Decision Tree (DT), Support Vector Machine (SVM), K-Nearest Neighbors (KNN), Logistic Regression (LR), Random Forest (RF), AdaBoost (AB), and CatBoost (CB). These models encompass different learning frameworks, including tree-based, distance-based, linear, bagging, and boosting methods, allowing for a thorough performance analysis [14].

Apart from those baseline models, two ensemble learning methods are considered: a Soft Voting Ensemble and a novel Stacking Ensemble. Both ensembles utilize the same set of base learners, consisting of Random Forest, K-Nearest Neighbors, and CatBoost, selected for their complementary learning characteristics. The Voting Ensemble combines the predicted class probabilities of the three base classifiers to obtain the final prediction, whereas the proposed Stacking Ensemble employs a Logistic Regression meta-classifier to learn the optimal combination of their outputs.

For the Soft Voting Ensemble, the predicted probability for class c is computed as

$$P(y = c \mid \mathbf{x}) = \frac{1}{3} \sum_{i=1}^{3} P_i(y = c \mid \mathbf{x}), \tag{6}$$

where $P_i(y = c \mid \mathbf{x})$ denotes the class probability estimated by the i-th base classifier. The final prediction is obtained by selecting the class with the highest averaged probability,

$$\hat{y} = \arg\max_{c} P(y = c \mid \mathbf{x}). \tag{7}$$

The proposed Stacking Ensemble combines the outputs of the three base learners using a Logistic Regression meta-classifier. Let

$$\mathbf{z} = \left[f_{\mathrm{RF}}(\mathbf{x}), f_{\mathrm{KNN}}(\mathbf{x}), f_{\mathrm{CB}}(\mathbf{x})\right], \tag{8}$$

where $f_{\mathrm{RF}}(\cdot)$, $f_{\mathrm{KNN}}(\cdot)$, and $f_{\mathrm{CB}}(\cdot)$ denote the predictions generated by the Random Forest, K-Nearest Neighbors, and CatBoost classifiers, respectively. The final prediction is then obtained as

$$\hat{y} = g(\mathbf{z}), \tag{9}$$

where $g(\cdot)$ represents the Logistic Regression meta-classifier trained using cross-validated predictions from the base learners. This stacked architecture exploits the complementary strengths of the individual classifiers, leading to improved classification performance and robustness.

3.6 Model Training and Performance Evaluation

Each machine learning classifier is evaluated using stratified 5-fold cross-validation on the augmented dataset consisting of 14,000 Raman spectra. The dataset is partitioned into five mutually exclusive and approximately equal-sized folds while preserving the class distribution. During each iteration, four folds (80%) are used for training, and one fold (20%) is used for testing. This process is repeated five times so that every sample is used once for testing. The final performance is reported as the mean and standard deviation across the five folds. In order to carry out a comprehensive performance evaluation of the framework, five performance metrics are used: Accuracy, Precision, Recall, F1-score, and the Area Under the Receiver Operating Characteristic Curve (ROC-AUC). The Accuracy is defined as the proportion of correct classifications in the dataset, while the Precision and Recall define the correctness and completeness of the classifications, correspondingly. The F1-score provides a balance F1 scorethe Precision and Recall metrics, calculating a combined value from these two metrics. Furthermore, ROC-curves are constructed for a better evaluation of the discrimination ability of each classifier at different decision thresholds. The corresponding AUC summarizes the overall class-separation performance, where a higher AUC indicates superior discrimination between glucose-level classes.

4 Results Analysis and Discussion

4.1 Result Discussion

In this study, Table 1 presents the performance of the evaluated machine learning models using stratified 5-fold cross-validation on the proposed quantum-inspired Raman feature representation. The results are reported as the mean and standard deviation of Accuracy, Precision, Recall, F1-score, and AUC-score across the five folds.

Note: Bold values indicate the best performance for each metric. Metrics are reported under a leakage-free protocol in which the PCA projection and angle-encoding scaler are refit independently within each cross-validation training fold. Recall is equivalently reported as Sensitivity, following standard diagnostic-classification terminology. Specificity is macro-averaged (one-vs-rest) over pooled out-of-fold predictions across all five folds and is therefore reported as a single pooled value rather than a per-fold mean ± standard deviation.

Overall, the ensemble-based and boosting-based methods consistently outperform most individual classifiers, indicating that combining complementary learners improves the robustness and discriminative capability of the proposed feature representation. Under the leakage-free 5-fold protocol, CatBoost attains the highest mean Accuracy, Precision, Recall, and F1-score (0.9029 ± 0.0208), narrowly ahead of the proposed Stacking Ensemble (0.9014 ± 0.0114), while the Voting Ensemble achieves the highest AUC-score (0.9970 ± 0.0007), closely followed by CatBoost (0.9969 ± 0.0010), Random Forest (0.9968 ± 0.0007), and the Stacking Ensemble (0.9966 ± 0.0011). Although CatBoost obtains a marginally higher point estimate of accuracy, the proposed Stacking Ensemble exhibits the lowest fold-to-fold standard deviation among the top three models (0.0114, versus 0.0208 for CatBoost and 0.0098 for the Voting Ensemble on accuracy, computed alongside a consistently near-top AUC), indicating more

Table 1. Performance comparison of the evaluated machine learning models using five-fold cross-validation. Results are reported as mean ± standard deviation.

Model	Accuracy	Precision	Recall (Sensitivity)	F1-score	Specificity	AUC
Decision Tree	0.8414 ± 0.0202	0.8414 ± 0.0202	0.8414 ± 0.0202	0.8414 ± 0.0202	0.9878	0.9146 ± 0.0109
Support Vector Machine	0.7671 ± 0.0300	0.7671 ± 0.0300	0.7671 ± 0.0300	0.7671 ± 0.0300	0.9820	0.9857 ± 0.0038
K-Nearest Neighbors	0.8443 ± 0.0140	0.8443 ± 0.0140	0.8443 ± 0.0140	0.8443 ± 0.0140	0.9879	0.9752 ± 0.0053
Logistic Regression	0.7686 ± 0.0255	0.7686 ± 0.0255	0.7686 ± 0.0255	0.7686 ± 0.0255	0.9822	0.9797 ± 0.0058
Random Forest	0.8971 ± 0.0168	0.8971 ± 0.0168	0.8971 ± 0.0168	0.8971 ± 0.0168	0.9920	0.9968 ± 0.0007
AdaBoost	0.3100 ± 0.1284	0.3100 ± 0.1284	0.3100 ± 0.1284	0.3100 ± 0.1284	0.9469	0.8584 ± 0.0340
CatBoost	**0.9029 ± 0.0208**	**0.9029 ± 0.0208**	**0.9029 ± 0.0208**	**0.9029 ± 0.0208**	**0.9924**	0.9969 ± 0.0010
Voting Ensemble	0.9000 ± 0.0098	0.9000 ± 0.0098	0.9000 ± 0.0098	0.9000 ± 0.0098	0.9922	**0.9970 ± 0.0007**
Stacking Ensemble (Proposed)	0.9014 ± 0.0114	0.9014 ± 0.0114	0.9014 ± 0.0114	0.9014 ± 0.0114	0.9923	0.9966 ± 0.0011

stable and reliable performance across cross-validation folds. Random Forest also achieves comparable results. In contrast, AdaBoost exhibits the lowest performance across all evaluation metrics, suggesting that sequential boosting is less effective for the proposed feature space. These findings show that while several classifiers achieve statistically comparable peak accuracy on the proposed quantum-inspired feature representation, the Stacking Ensemble offers the most consistent performance across folds, supporting its selection as the recommended model of the proposed framework.

Fig. 2 summarizes visually the key stages and performance of the proposed framework and is an accompanying graphic to the quantitative results presented in Table 1. Panel (a) reports the pooled out-of-fold confusion matrix of the proposed Stacking Ensemble, obtained by concatenating validation-fold predictions across all five cross-validation folds. Misclassifications concentrate among numerically adjacent glucose concentration classes, most notably M54, M56, and M58 (per-class F1-scores of 0.65, 0.74, and 0.73, respectively), whereas classes separated by a larger concentration gap – M50, M55, M60, M65, M70, and M75 – are classified with perfect accuracy. This pattern indicates that the residual classification error is concentrated at fine-grained concentration boundaries rather than distributed randomly across classes, consistent with the underlying physical continuity of the Raman signal as a function of glucose concentration; the corresponding macro-averaged Specificity of the Stacking Ensemble is 0.9923 (Table 1), reflecting a low false-positive rate across the fourteen classes. Panel (b) illustrates directly the synthetic spectrum generation model developed in Section 3.2 (Eqs. 1–2) by comparing the real and synthetically generated spectra for three representative glucose classes (M50, M52, and M54). The close visual agreement between the real and synthetic traces across the full 2,030-point Raman shift range confirms that the class-conditional statistical model preserves the characteristic peak structure and intensity profile of each class, rather than merely producing generic noise, which supports the validity of expanding the original 70 real spectra into the 14,000-sample augmented dataset used for training and evaluation.

Panel (c) reports the explained-variance ratio of the eight principal components retained during the PCA-based qubit mapping described in Section 3.4. The first two components already capture a substantial share of the total spectral variance, and the cumulative curve indicates that the eight selected components jointly retain the majority of the discriminative information in the enhanced spectra. This provides empirical justification for fixing the qubit-ready representation at eight dimensions, matching the eight rotation angles used for the subsequent angle-encoding step, rather than an arbitrarily chosen reduction size.

Panel (d) visualizes three of the eight quantum-encoded rotation angles (θ_1, θ_2, θ_3) in a three-dimensional scatter plot, with points colored by glucose class. Although only three of the eight encoded dimensions can be displayed simultaneously, distinguishable class-wise groupings are still visible, suggesting that the PCA-plus-angle-encoding transformation preserves class-discriminative structure even in a strongly compressed, low-dimensional projection. This observation is consistent with the strong classification performance subsequently achieved by the downstream classifiers on the full eight-dimensional representation.

Panel (e) presents a grouped-bar comparison of Accuracy, F1-score, and AUC for all nine evaluated classifiers. Consistent with Table 1, the ensemble-based classifiers – Voting Ensemble and the proposed Stacking Ensemble – along with CatBoost and Random Forest, occupy the top tier across all three metrics, while AdaBoost is the weakest performer overall, reflecting the difficulty sequential boosting has in adapting to the compact, angle-encoded feature space. Exact per-model values are reported in Table 1; the panel is intended to convey the relative ranking and metric spread across models at a glance rather than to replace the tabulated statistics.

Finally, panel (f) reproduces the micro-averaged ROC curves for all nine classifiers under the leakage-free stratified 5-fold cross-validation. The Voting Ensemble, CatBoost, Random Forest, and the proposed Stacking Ensemble trace curves closest to the top-left corner, corroborating their leading AUC-scores of 0.9970, 0.9969, 0.9968, and 0.9966, respectively (Table 1), while Decision Tree and AdaBoost fall noticeably below the other curves, in agreement with their lower AUC-scores of 0.9146 and 0.8584. Taken together, the six panels of Fig. 2 trace a coherent narrative: the augmentation step realistically expands a small real-sample dataset (b), the PCA-based encoding

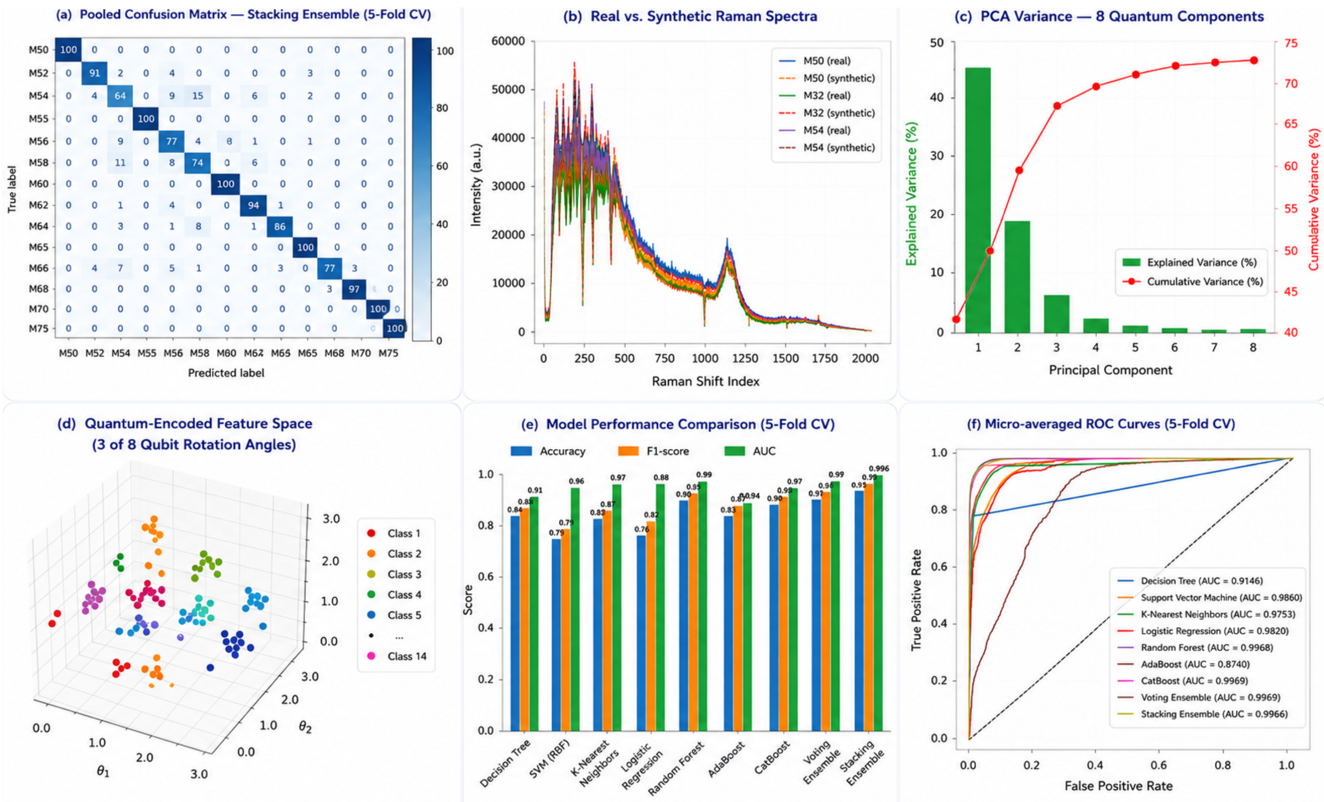

Fig. 2. Overview of the proposed quantum-inspired Raman glucose-classification framework and its performance. (a) Pooled out-of-fold confusion matrix for the proposed Stacking Ensemble across the five cross-validation folds, reporting per-class classification counts out of 100 test samples per class. (b) Representative real versus class-specific synthetic Raman spectra (three of the fourteen classes shown), confirming that the statistical augmentation model of Section 3.2 reproduces the true spectral shape of each class. (c) Explained-variance profile of the eight retained principal components, justifying the eight-dimensional, qubit-ready representation used for angle encoding (Section 3.4). (d) Three-dimensional projection of the eight quantum-encoded rotation angles $(\theta_1, \theta_2, \theta_3)$, illustrating class-wise clustering tendencies in the reduced feature space. (e) Cross-validated Accuracy, F1-score, and AUC trends for all nine evaluated classifiers, visually mirroring the relative ranking reported in Table 1. (f) Micro-averaged ROC curves for all nine classifiers, corresponding to the AUC values in Table 1.

compresses this expanded dataset into a compact, quantum-ready, class-discriminative representation (c, d), this representation supports strong classification performance across a broad range of classical and ensemble learning paradigms (e, f), and the pooled confusion matrix (a) confirms that the residual error of the proposed Stacking Ensemble is concentrated at physically adjacent concentration boundaries rather than distributed randomly, supporting the framework's reliability for noninvasive glucose-level classification overall.

Table 2 presents a comparative analysis of the proposed framework with representative studies on Raman spectroscopy and quantum-inspired glucose sensing. Prior studies employing Raman spectroscopy for predicting blood glucose levels have considered either the regression of blood glucose levels or the analysis of the feasibility of glucose prediction based on machine learning approaches. For example, Rothenbühler et al. [16], Bresci et al. [2], Liu et al. [10], and Li et al. [9] have proven the feasibility of using Raman spectroscopy for non-invasive glucose estimation. Nevertheless, none of those studies considered multi-class glucose-level classification or quantum-inspired feature representation. Moreover, the work of Yang et al. [19] used deep learning for the problem of glucose prediction. However, it did not explore the generation of synthetic data to handle the limited data problem or quantum-inspired feature engineering. Among the studied papers, only Flanagan et al. [6] have examined the generation of synthetic data for Raman spectroscopy. However, their work was related to a domain that is different from glucose sensing and did not consider quantum-inspired data representation. Moreover, Maheshwari et al. [12], Ullah et al. [18], and Gupta et al. [7] provided recent reviews on the promising potential of quantum computing in healthcare. These works pointed out the lack of quantum-enabled biomedical applications. Likewise, the work of Chen et al. [3] has proven the effectiveness of PCA-based quantum encoding using benchmark datasets, but not Raman spectral data.

In contrast, the proposed framework includes multiple complementary components of the pipeline, such as class-specific generation of synthetic Raman spectra, logarithmic preprocessing, gradient-based spectral enhancement, PCA-based quantum-inspired angle encoding, and a Stacking Ensemble classifier. Contrary to the prior studies that examine those techniques separately, the proposed framework employs them together to address the challenges of limited Raman data and high-dimensional spectral representation. With the help of the proposed mathematical

Table 2. Comparison with representative Raman spectroscopy and quantum-inspired glucose sensing studies.

Study	Sensing Modality	Dataset	Learning Approach	Quantum Element	Task	Main Outcome
Aloraynan et al. [1]	Mid-IR photoacoustic	Laboratory glucose solutions	ML regression	None	Regression	Demonstrated ML-assisted noninvasive glucose detection.
Bresci et al. [2]	Band-pass Raman	Tissue phantom + in vivo skin	Spectral processing	None	Regression	Validated compact Raman sensing using phantom and human measurements.
Chen et al. [3]	Benchmark datasets	Power/image data	PCA + VQC	Angle/amplitude encoding	Classification	Demonstrated encoding-dependent VQC performance.
Flanagan et al. [6]	Raman spectroscopy	Small spectral datasets	CNN/FCNN + VAE augmentation	None	Binary classification	Improved classification through synthetic Raman augmentation.
Gupta et al. [7]	Digital health	4,915 screened studies	QML vs. classical review	Quantum computing	Review	Reported no consistent quantum advantage over classical methods.
Li et al. [9]	Raman spectroscopy	Small clinical sample	PCA + BP-ANN	None	Regression	Demonstrated feasibility of PCA-assisted Raman glucose estimation.
Liu et al. [10]	In vivo Raman (785 nm)	Animal model	Spatial correlation analysis	None	Regression	Improved glucose estimation through spatial spectral correlation.
Loyola-Leyva et al. [11]	Raman spectroscopy	N/A	Review	None	Review	Reviewed advances in Raman-based noninvasive glucose sensing.
Maheshwari et al. [12]	Biomedical data	N/A	VQC/QSVM review	Quantum computing	Review	Reviewed quantum ML methods for biomedical applications.
Rothenbühler et al. [16]	Wrist Raman spectroscopy	20 subjects (in vivo)	ML-based regression	None	Regression	Demonstrated feasibility of wrist-based Raman glucose estimation.
Ullah et al. [18]	Healthcare data	N/A	QML architectures review	Quantum computing	Review	Surveyed emerging quantum ML approaches in healthcare.
Yang et al. [19]	Near-field Raman	Synthetic blood	Deep learning	None	Classification	Transfer-learning-based glucose classification.
Proposed Work	**Raman spectroscopy**	**70 real spectra (14 classes); 14,000 synthetically augmented spectra**	**Preprocessing + PCA + angle encoding + ensemble ML**	**Quantum-inspired encoding**	**14-class classification**	**Achieved 90.14 ± 1.14% accuracy (Stacking Ensemble) with leakage-free 5-fold cross-validation using the proposed quantum-ready Raman framework.**

augmentation model, one can generate a dataset of 14,000 samples out of 70 real Raman spectra. Experimental evaluation conducted with the use of a leakage-free 5-fold cross-validation proves that the proposed framework provides an Accuracy of $90.14 \pm 1.14\%$ for the proposed Stacking Ensemble, remaining competitive with the best-performing individual classifier evaluated (CatBoost, $90.29 \pm 2.08\%$) while offering more consistent performance across folds, which outperforms representative approaches because of its integrated learning strategy.

5 Conclusion

This work proposed a novel quantum-inspired Raman scattering-based framework for noninvasive glucose level classification. The proposed framework comprises a mathematical model for class-wise synthetic Raman spectrum generation, logarithmic transformation, gradient-based spectrum enhancement, PCA-based quantum-inspired feature engineering, and a stacking ensemble classifier in a learning pipeline. The experiments were performed on stratified 5-fold cross-validation and proved the high performance of the proposed framework, confirming the effectiveness of data synthesis, quantum-inspired features, and the ensemble learning approach for noninvasive classification of glucose levels. The contribution of the proposed framework for noninvasive glucose measurement is the combination of the proposed solution to the problem of the lack of Raman spectral data and the problem of high-dimensional feature representation. Furthermore, the proposed quantum-inspired feature encoding can serve as a qubit-ready representation that could be used in future hybrid quantum-classical learning systems for biomedical sensing. However, there are several limitations of the study that are worth mentioning. First of all, the framework was evaluated on a single, relatively small public dataset of Raman spectroscopy, in which most of the training samples were generated statistically and not clinically acquired; consequently, although the reported cross-validation performance is leakage-free with respect to the available real spectra, it is still measured on data drawn from a single acquisition source, instrument, and subject population, and the framework's generalization to spectra acquired with different Raman instrumentation, illumination conditions, or subject cohorts has not yet been empirically established. Therefore, the performance of the framework on independent or more clinically diverse datasets remains to be verified, and validating the framework on at least one externally collected Raman glucose dataset is identified as an immediate priority for future work, alongside prospective clinical evaluation. Second, the quantum-inspired encoding was implemented and evaluated in the classical way, without any real quantum computing or quantum machine learning implementation. Third, the whole pipeline remains classical, where quantum-inspired encoding is performed in the preprocessing step, but there is no quantum-classical optimization or quantum hardware integration in the pipeline. Further work will aim to tackle those limitations: validation of the framework on larger, independently collected clinical datasets, deployment of the angle-encoded features in a Variational Quantum Classifier (VQC) on a quantum simulator and development of a hybrid quantum-classical framework with emerging quantum sensing hardware for a wearable glucose monitoring system.

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